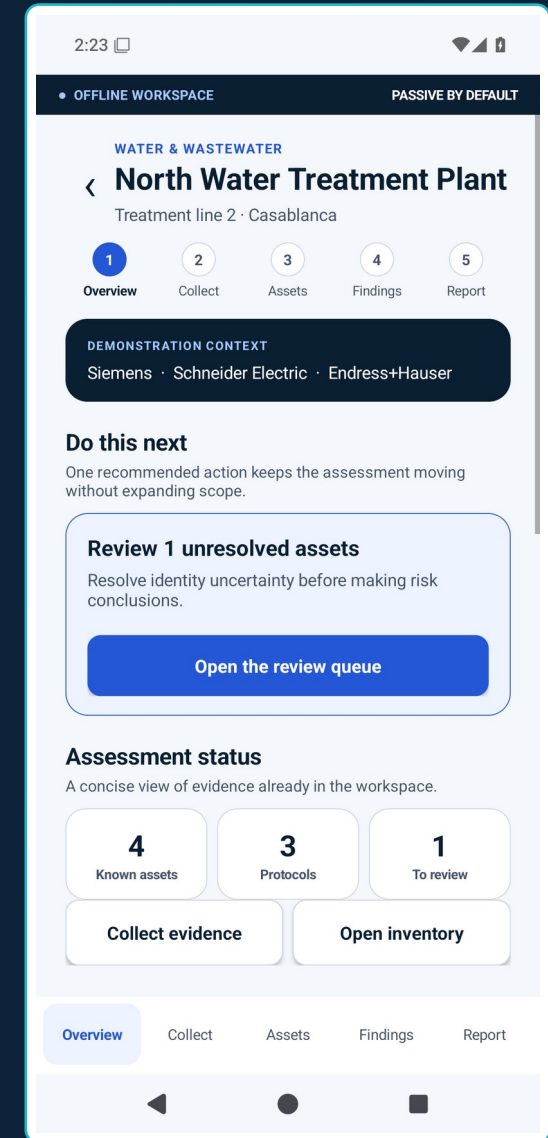


Know what changed before you sign off.

Turn a bounded water-OT evidence window into a reviewable inventory, explicit findings and a controlled handoff - without pretending scanner output is truth.

▶ [WATCH THE ACTUAL USER STORY](#)

Continuous Android 15 screen recording. Same application, same user story.



Actual sample water-treatment workspace

The costly failure is signing off with uncertainty.

A useful field instrument closes decision gaps instead of maximizing device counts.

At the end of a site visit, an authorized assessor still has to answer:

01

What changed?

Which identities differ from baseline or remain unknown?

02

What can I defend?

Which claims retain evidence, provenance and confidence?

03

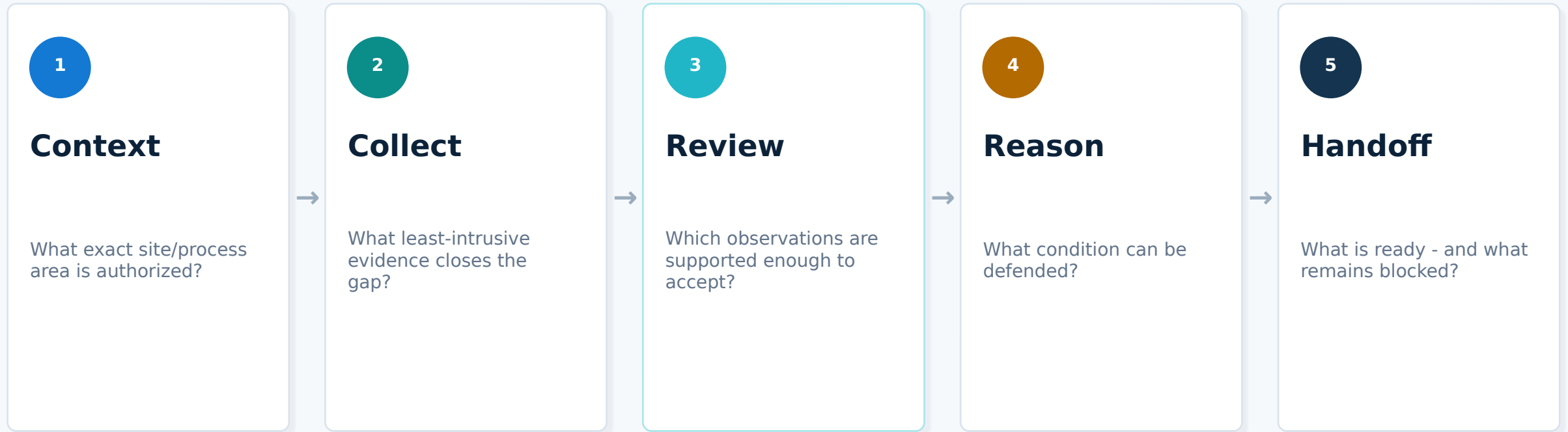
What is safe to claim?

Which conclusions remain blocked by visibility or approval gaps?

A useful answer can be “not enough evidence yet” - as long as the missing evidence is explicit.

One user story. Five decisions.

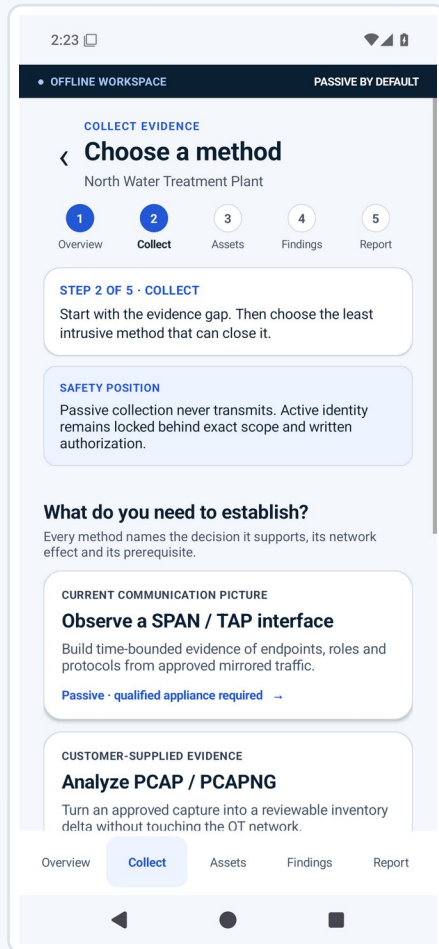
The deck and video now follow the assessor workflow in sequence.



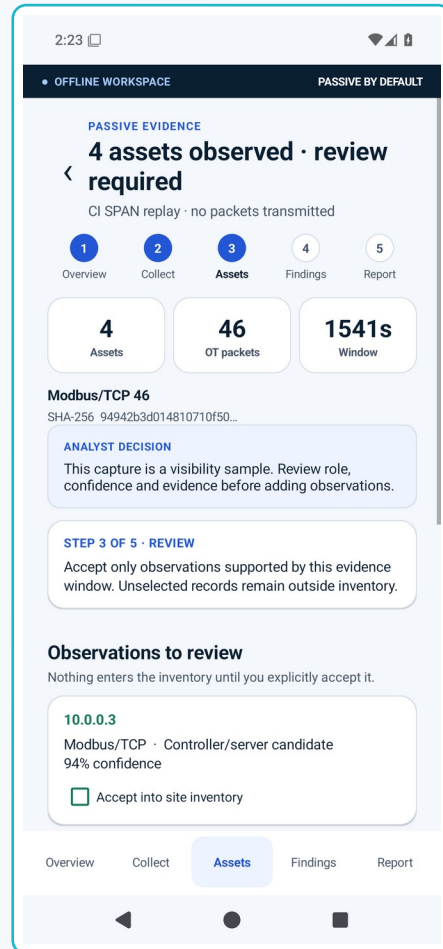
The user story is the product: every screen exists to support one decision.

Passive evidence is not inventory until an analyst accepts it.

This is the core trust behavior: observations enter a review queue before they change the model.



Choose least-intrusive method



Review and accept observations

USER STORY

“I need a current communication picture without touching the process.”

- Choose SPAN/TAP or an approved capture before active methods.
- Hash and parse the bounded evidence window.
- Accept only the observations the analyst is prepared to defend.

No acceptance → no inventory mutation.

Active behavior fails closed before a packet is sent.

The demo deliberately asks for an out-of-scope target first.

2:23

AUTHORIZED ACTIVE CHECK

< Identify one Modbus device

North Water Treatment Plant

1 Work order

Case reference

P0-WATER-001

Process area

Treatment line 2

2 Exact target and scope

Target controller IPv4

10.0.2.2

Authorized IPv4 scope (CIDR)

10.0.2.0/24

Modbus unit ID

1

ACTIVE LIMITS

1 identity request · TCP/502 · 1.5 s timeout · no register reads or writes

3 Authorization

☐ I confirm written operational and security authorization for this target, scope and time window.

Exact work order, target and CIDR
Emulated Android evidence - not physical field qualification

2:23

AUTHORIZED ACTIVE CHECK

< Identify one Modbus device

North Water Treatment Plant

1 Work order

Case reference

P0-WATER-001

Process area

Treatment line 2

2 Exact target and scope

Target controller IPv4

192.0.2.5

Authorized IPv4 scope (CIDR)

10.0.2.0/24

Modbus unit ID

1

ACTIVE LIMITS

1 identity request · TCP/502 · 1.5 s timeout · no register reads or writes

3 Authorization

☒ I confirm written operational and security authorization for this target, scope and time window.

Target is outside the authorized CIDR. The app will not expand

Local scope rejection

FAIL-CLOSED

192.0.2.5

is rejected against

10.0.2.0/24

- Validation happens in the Case App before broker execution.
- The application does not broaden scope to make a test pass.

Then ask one exact identity question - and stop.

The controlled CI target answers a bounded Modbus basic-device-identification request.

10.0.2.2

EXACT TARGET

FC 43 / 14

ONE OPERATION

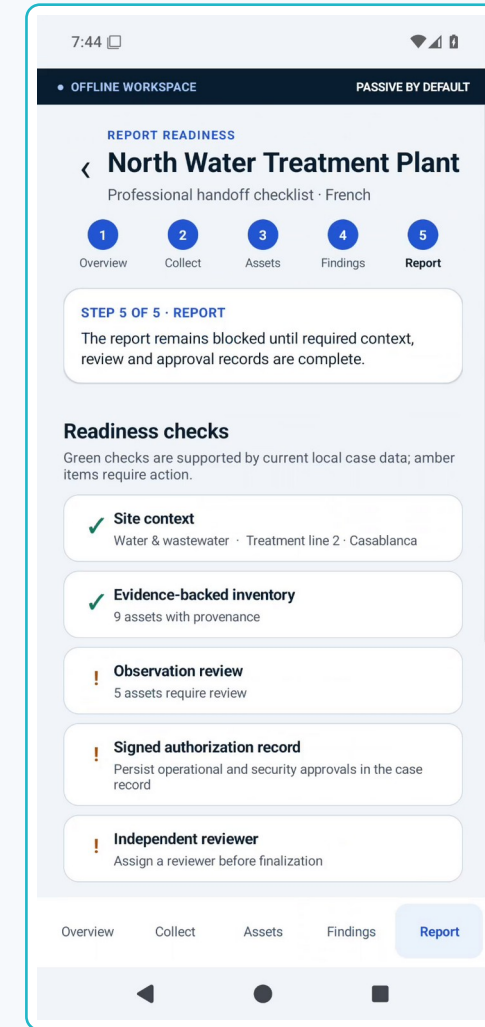
0

REGISTER READS / WRITES

CONTROLLED ACTIVE PATH

Case App → signed one-use grant → Network Broker → TCP/502 → PyModbus

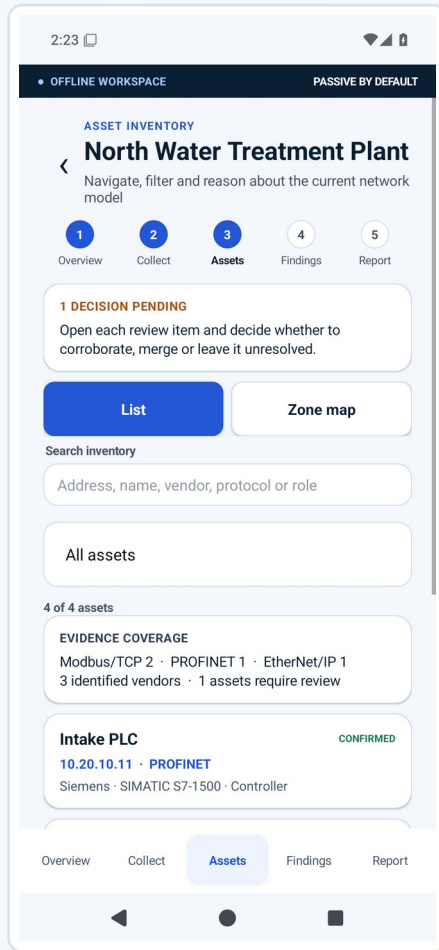
The result carries the evidence needed to decide whether the identity can be accepted into inventory.



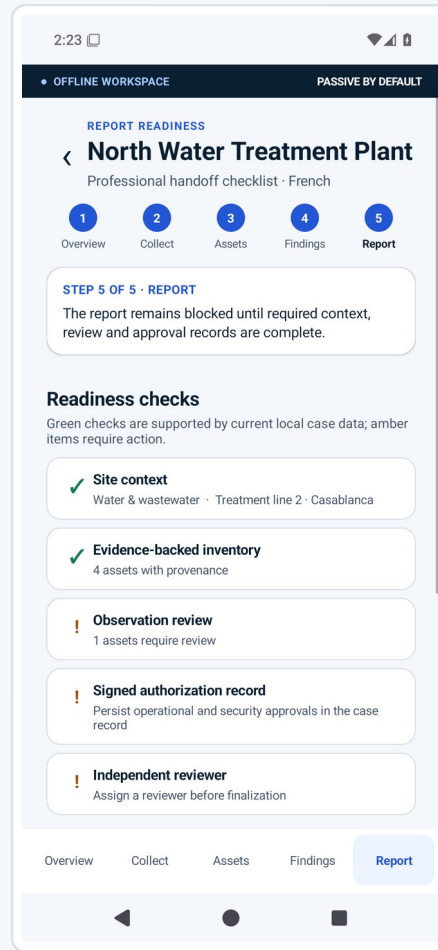
Actual PyModbus CI identity result

From observations to a defensible handoff.

The product preserves uncertainty instead of hiding it.



Evidence-backed inventory



Explicit report blockers

Customer-facing outcome

- Inventory delta** Corroborated, unexpected, missing-from-evidence and unresolved identities.
- Provenance** What was observed, where it came from and confidence.
- Finding drafts** Condition, evidence, context and required validation.
- Readiness blockers** Authorization and independent review remain visible.

A blocked report is useful when the evidence is not ready.

Built for a Moroccan water-utility pilot - not generic industry.

The commercial scope now matches P0-WATER: drinking water and wastewater operations only.

Water operating contexts

Treatment

process/control

Pumping

stations/boosters

Reservoirs

storage/transfer

Telemetry

remote monitoring

District metering

flow/pressure

Leak reduction

instrumentation

Wastewater

collection/treatment

BEACHHEAD

ONEE - Water Branch

National water production, transmission and sanitation programs.

Regional multiservice companies

Only their drinking-water and liquid-sanitation responsibilities are in scope.

Entry offer: one bounded, passive-first evidence baseline.

What is proven today - and what is not.

Credibility comes from stating the proof boundary before the buyer asks.

PROVEN IN CI / EMULATION

- Android API 29 + API 35 guided journeys
- PCAP / PCAPNG passive analysis with provenance
- Receive-only capture boundary exercised in CI emulation
- One signed Modbus basic identity operation
- Interoperability: PyModbus, modbus-tk and Conpot
- Review-first inventory, findings and readiness UI

NOT YET QUALIFIED

- Physical Samsung / custom appliance image
- USB-Ethernet + physical SPAN/TAP compatibility
- Packet loss and behavior at production rates
- Real PLC firmware and production OT networks
- Encrypted multi-user case vault + reviewer signature
- Deterministic signed final report package

The pilot exists to close the physical-field qualification gap.

Pilot one bounded water segment.

Not a plant-wide scan. Not compliance certification. A witnessed test of whether the evidence pack helps an authorized assessor make a better decision.

1 Scope

Name one treatment, pumping or wastewater control segment.

2 Baseline

Use approved passive evidence first.

3 Close a gap

Run one exact active identity request only when needed.

4 Review

Judge inventory delta, evidence quality and remaining blockers.

▶ [WATCH THE ACTUAL USER STORY](#)

Decision after the pilot

Is the evidence good enough to justify physical hardware qualification and a controlled field evaluation?